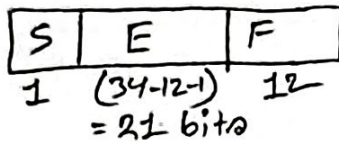


1. Given the number $A = 1.1011 \times 2^{232500}$; can this number be represented using a 34-bit IEEE-754 floating-point format, where the fraction field is 12 bits?

You must provide a justification for your answer.

[2+1]

Answer:



$$\text{Bias} = 2^{21-1}$$

$$= 1048575$$

Biased Exp

$$= 1048575 + 232500$$

$$= 1281075$$

$$\text{Biased Exp. Range} = 0 \text{ to } 2^{21}-1$$

$$= 0 \text{ to } 2097151$$

$$= 1 \text{ to } 2097150 \text{ [usable]}$$

$$1281075 < 2097150$$

$\therefore$ The number can be represented.

2. Suppose you have two floating point numbers 3.14 and 2.15 stored respectively in f2 and f3 registers. Now you run the following code:

feq.s X3, f2, f3

After running the code, what would be the value inside the X3 register?

Note: both values stored in f2 and f3 are in IEEE-754 single precision format. [1]

Answer: $f_2 \neq f_3$; So, $X_3 = 0$

3. Given two numbers 119_{10} and 127_{10} . We want to multiply these two numbers using the Long Multiplication method. Answer the following questions: [4]

$$119 = \boxed{1110111}^{7 \text{ bits}} \quad | \quad 127 = \boxed{1111111}^{7 \text{ bits}}$$

- How many iterations are necessary to complete this multiplication?
- What would be the size of the product register?
- What would be the size of the multiplier register?
- What would be the size of the inputs of ALU?

Answer:

- 7 bits
- 14 bits
- 7 bits
- 14 bits

4. Given a number, $W = 0xA00A05002$ in 36-bit IEEE-754 representation, where the size of the fraction field is 16 bits. Find the equivalent decimal number of the given number W . [7]

$A00A05002$

$= \underbrace{1010\ 0000\ 0000\ 1010\ 0000}_{\text{exponent}} \underbrace{0101\ 0000\ 0000\ 0010}_{\text{fraction}}$

$$\text{Bias} = 2^{19-1} - 1 = 262143$$

$$\text{Biased Exponent} = 010\ 0000\ 0000\ 1010\ 0000 \\ = 131232$$

$$\therefore \text{Actual Exponent} = 131232 - 262143 \\ = -130911$$

$$\Rightarrow \text{Decimal number} = (-1)^1 \times (1 + 0.0101\ 0000\ 0000\ 0010) \times 2^{-130911} \\ = -1.0101\ 0000\ 0000\ 0010 \times 2^{-130911}$$

The exponent is too ~~small~~ small so, calculator won't be able to convert it.